



DID YOU KNOW – ATC – AUTOMATIC TUBE COMPENSATION

Dräger



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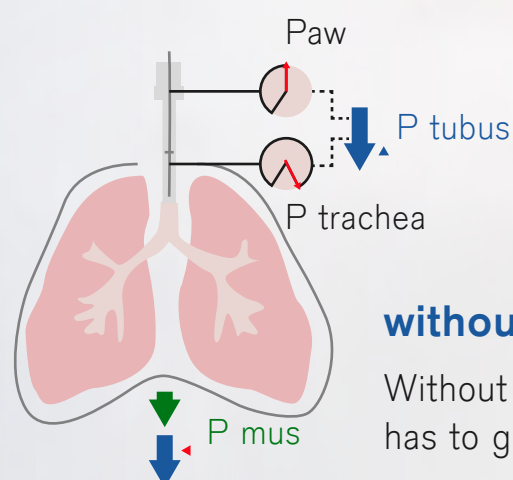
• What is it?

ATC stands for Automatic Tube Compensation and it compensates for the artificial airway resistance (endotracheal tube or tracheostomy).

The objective is to give the patient the feeling that, in terms of work of breathing, he's not intubated.

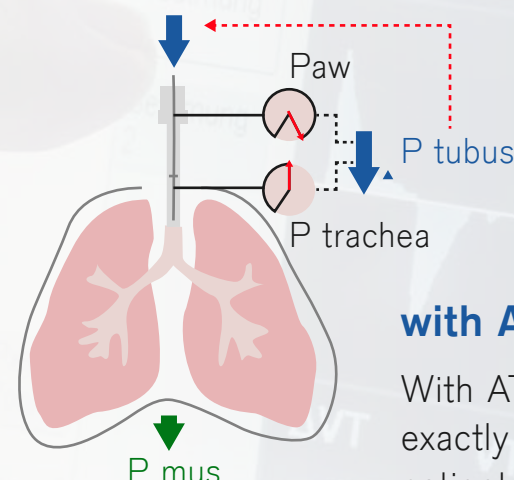
ATC is an addition to ventilation modes in Dräger ventilators.

How is it working?



without ATC

Without ATC, the patient has to generate ΔP_{tube} .



with ATC

With ATC, the ventilator produces exactly this ΔP_{tube} and relieves the patient of the extra work.

• Automatic Tube Compensation settings (in V-family)

ATC is available in the “Advanced settings” tab.

Settings for Expiratory and Inspiratory compensation can be made in System setup.

Switch on Automatic Tube Compensation – 1

2 Calculation of tracheal pressure – Chose if you want to see the calculation of tracheal pressure P_{trach} (green curve)

3 Tube type – Set the Tube type (endotracheal or tracheostomy)

4 Tube diameter mm – Set the tube diameter

5 100 Compens – Set the % of compensation (0 = no compensation, 100 = full compensation)

Overview	AutoFlow	Trigger	Sigh	Apnea ventilation	ATC
Automatic tube compensation	On	Off			
Calculation of tracheal pressure	On	Off			
Tube type	Endo-trach.	Trach.			
Tube diameter mm	8.0				
Exp. compensation	On				
Insp. compensation	Spont. + mand.				

Why is it helpful to improve outcome?

The spontaneously breathing intubated patient has to perform work of breathing to overcome the tube resistance [1]

ATC takes over the work of breathing induced by the tube resistance

ATC provides adequate flow-dependent ventilatory support

The patient breaths like without any tube [1]

References:

[1] - Respiratory comfort of automatic tube compensation and inspiratory pressure support in conscious humans; Guttman, J. et al., Intensive Care Medicine 1997, Vol. 23, No.11, 1119-1124